Unit 1 - Supervised Learning

What is Machine Learning?

Machine Learning is a subset of AI that enables computers to learn from data and improve their performance on tasks without being explicitly programmed. Instead of writing rules manually, ML algorithms discover patterns in data automatically.

Types of Machine Learning

Supervised Learning

Trains on labeled data where input-output pairs are known. The model learns to map inputs to outputs. Examples: Classification, Regression.

Unsupervised Learning

Trains on unlabeled data and finds hidden patterns or structure. Examples: Clustering, Dimensionality Reduction.

Reinforcement Learning

Agent learns by interacting with environment, receiving rewards or penalties. Examples: Game playing AI, Robotics.

Supervised Learning Algorithms

- Linear Regression - predicts continuous values
- Logistic Regression - binary classification
- Decision Trees - tree-based classification and regression
- Random Forest - ensemble of decision trees
- Support Vector Machine (SVM) - finds optimal hyperplane
- K-Nearest Neighbors (KNN) - classifies based on proximity
- Naive Bayes - probabilistic classifier

Model Evaluation Metrics

Accuracy

Percentage of correct predictions. Best for balanced datasets.

Precision and Recall

Precision = TP/(TP+FP). Recall = TP/(TP+FN). Used for imbalanced datasets.

F1 Score

Harmonic mean of Precision and Recall. F1 = 2*(P*R)/(P+R).

Mean Squared Error

Used for regression problems. Average of squared differences between predicted and actual values.

Overfitting and Underfitting

Overfitting occurs when a model learns training data too well including noise, performing poorly on new data. Underfitting occurs when a model is too simple to capture the underlying patterns. The goal is to achieve the right balance through techniques like cross-validation, regularization, and proper model selection.